

MODULAR BLUEPRINT OF PHYSICAL AGENT ARCHITECTURE

Layered Architectural Contract: Observations, World Models, Intent Leases, Spline Chunks, and Barrier Reflexes

LAYER 1: SENSORY CONTRACT

OBS-01: Zero-Copy Ring Buffers & Hardware Timestamps

LAYER 2: SPATIAL STATE CONTRACT

STATE-01: Dynamic SE(3) Coordinate Frame Tree & Proprioceptive Innovation

LAYER 3: INTENT LEASE CONTRACT

INTENT-01: 3D Workspace Grounding & Monotonic Countdown Leases (t_{expire})

LAYER 4: TRAJECTORY CHUNK CONTRACT

PLAN-01: $H=16$ Step Action Chunks & C^2 Quintic Spline Continuity

LAYER 5: SAFETY ENFORCER CONTRACT

ENF-01: 1 kHz Zero-Malloc Active-Set QP Barrier Filter ($h(x) \geq 0$)